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Education and crime: recent evidence from Italian provinces

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Gratitude to Prof. Anelli and colleagues Francesco and Guido. A mention to Andrea and Ilaria, who fortuitously sparked my curiosity on this topic.

Then, a mention to my girlfriend Angela and to all the friends encountered here at Bocconi. Finally, to my mum Roberta and the rest of the family.

Education and crime: recent evidence from Italian provinces

by Tommaso Piglicampo

Abstract

This work explores the relationship between education and crime taking advantage of aggregate data from the last decade for Italian provinces. A brief yet comprehensive review of the literature on the topic initially points out which factors affect crime and then deepens the role of education. Various metrics regarding standardized test scores (INVALSI) are merged with crime statistics from the Italian Ministry of Interior into a panel. The impact of education on crime rates is estimated through weighted least squares (WLS) regression. A robust negative correlation is found between test scores and the number of criminals per capita, although no significant relationship between test scores and total crime rates emerges. A general downward trend in the number of criminals goes along significative differences between provinces. While the results are robust to endogeneity, multicollinearity between education and other factors affecting crime is present.

Table of Contents

1. Introduction

Crime imposes a rather substantial burden on society. Jens Ludwig, a Professor at Georgetown University, estimated this cost at $2 trillion annually in a 2006 statement.[1] Therefore, understanding criminal behavior is not merely an academic exercise but has substantial policy implications. Concurrently, criminals tend to be significantly less educated than the general population.[2]

While most policies focus on the incapacitation or rehabilitation of offenders to address the cost of crime, the concentration of crime among few perpetrators offers a unique opportunity for policy interventions to make a substantial impact if they can prevent the development of future criminals.[3] Estimates from Anders, Barr and Smith (2023) suggest that preventing the development of a single career criminal could result in more than 100 fewer annual victims of crime. A stitch in time may save nine.

This work aims to contribute to the existing literature on the topic of education and crime not only by providing a brief yet profound review, but also to take advantage of very recent and broad, yet granular, data. Moreover, the literature on the topic focuses either on the quantity or the quality of schooling, rather than on the student's outcome, which is the instrument for education used in this work, indeed.

Dataset is a panel built along Italian provinces. Data regarding education results from standardized tests that all Italian students must assess during

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[1] As reported in Anders, Barr and Smith (2023), at the time of the statement (2006) $2 trillion were roughly 17 percent of the annual US GDP (*The Costs of Crime, before the United States Senate Committee on the Judiciary, 107th Cong.* Statement of Jens Ludwig, Professor, Georgetown Public Policy Institute, Georgetown University).

[2] Among others, see Harlow (2003) for an insight on the US situation. Machin, Marie and Vujić (2011) and Hjalmarsson, Holmlund and Lindquist (2015) saw this pattern in UK and Sweden too. About Italy – the country on which this work focuses – Buonanno and Leonida (2006) reported that in 2001 more than 75% of Italian convicts had not completed high school.

[3] As Anders, Barr and Smith (2023) pointed out, Farrington (2006) mentioned that less than 6% of the population commit most crimes.

compulsory education. They are linked with administrative data regarding the number of crimes and criminals from the Department of Public Safety of the Italian Ministry of Interior.

No “natural experiment” setting has been identified. The analysis explores the linear relationship between education and crime using weighted least squares (WLS) regression, with the population of provinces as weights. A consistent negative association is observed between test scores and the number of criminals across various datasets. The association remains robust when alternative measures of educational attainment are used. In contrast, no relationship has been found between test scores and the total number of crimes. Finally, multicollinearity represents an obstacle when controls (in accordance with existing literature) are included in the model.

The remainder of this work is structured as follows: Section 2 consists of a literature review on crime, divided into two subsections. The first subsection explores its relationship with various factors and the second deepens its connection with education. Section 3 describes the data used in the analysis. Section 4 details the empirical strategy. Section 5 presents the results and finally, Section 6 offers the conclusion. References and figures can be found afterwards, in Sections 7 and 8, respectively.

2. Literature review

2.1 Factors related to crime

2.1.a Deterrence, punishment and incarceration

The earliest strand of economic research on crime, beginning with Becker (1968), regards deterrence and punishment as the mechanism to contrast crime (e.g. Ehrlich, 1975; Witte, 1980; Tauchen, Witte and Griesinger, 1994). Levitt (1998) confirmed the substantial role of deterrence but did not find that the punitiveness of the juvenile justice system is impactful. Results from Corman and Mocan (2000) also supported the deterrence hypothesis, showing that increases in arrests lead to declines in murders, robberies, burglaries and motor vehicle thefts. A larger police force also reduces robberies and burglaries. Similarly, Kelly (2000) noticed that police activity has relevant effects on property crime. Regarding incarceration, Needels (1996) reported high recidivism rates among prisoners, a finding later confirmed by Aizer and Doyle (2015), who found that juvenile incarceration results in higher adult incarceration rates and significantly lower high school completion rates – hence negatively affecting human capital accumulation. On one hand, Freeman (1996) estimated that even marginally effective prevention policies might be worth the cost of incarceration from the perspective of society. On the other hand, he argued that the depressed labor market for low-skill American workers was a key factor contributing to the persistently high levels of crime among less-educated men, regardless of the deterrence mechanisms in place.

2.1.b Economic factors

Economic factors have been examined for their impact on crime. Grogger (1998) connected crime to wages, concluding that juvenile crime responds to price incentives and that declining real wages may have contributed to the rise in youth crime during the 1970s and 1980s. Moreover, he argued that wages largely explain – as they rise with age - the well-known tendency for crime to decrease with age. Kelly (2000) found that poverty has significant effects on property crime, but little on violent crime. Machin and Meghir (2004) looked at cross-area changes in crime and the low-wage labor market in England and Wales, finding that crime decreased in areas where wage growth in the poorest quartile of the distribution was faster. Bell, Bindler and Machin (2018) demonstrated that graduating high school during a recession, along with numerous other negative outcomes cited in their literature review, can significantly increase the affected cohorts' arrest rates well into adulthood. Conversely, Fajnzylber, Lederman and Loayza (2002b) noted that not only property crime, but also violent one decreases when economic growth improves. Quite counterintuitively at first glance but aligned with all those studies mentioned above, they also did not find any relationship between the mean level of income and crime. Overall, it appears that crime tends to be counter-cyclical (Fajnzylber, Lederman and Loayza, 2002a). Moreover, it is important to note that a positive correlation between property crime and income inequality was observed quite some time ago (e.g., Ehrlich, 1973). More recently, Kelly (2000) observed that inequality has no effect on property crime but a strong and robust impact on violent crime. Fajnzylber, Lederman and Loayza (2002b) found a worldwide causal relationship between income inequality and both property and violent crime.

2.1.c Social factors

Among social factors, Patacchini and Zenou (2012) found that conformity plays a vital role in all crimes, especially petty crimes. The number of female-headed households has also been mentioned as a strong predictor of city crime rates (Glaeser, Sacerdote and Scheinkman, 1996). Donohue and Levitt's (2001) findings suggest that legalized abortion has resulted in a significant drop in crime, which indirectly supports the significance of parental variables in lowering crime rates. Font and Kennedy (2022) produced an extensive literature review on childhood neglect, arguing that it represents a critical yet understudied context in crime and delinquency. They highlighted that childhood neglect influences crime both across generations (e.g., maltreated children are at a higher risk of delinquency and crime) and within generations (e.g., parents who engage in criminal behavior are more likely to abuse or neglect their children and expose them to abuse by others).

Regarding the relationship between disreputable neighborhoods and crime, Sah (1991) presented a model where an individual's decision to become a criminal decreases the likelihood of any other individual being arrested. Murphy, Snleifer, and Vishny (1993) argued that as the number of criminals increases, the returns from legal activities decline because criminals steal legal revenues. Among many scholars who addressed the topic,[4] Case and Katz (1991) found evidence consistent with neighborhood effects in low-income neighborhoods in Boston. Using a different approach, Glaeser, Sacerdote and Scheinkman (1996) reported similar findings. [5] Ludwig, Duncan and Hirschfield (2001) utilized the outcome of the Moving to Opportunity project to notice that providing families with the opportunity to move to lower-poverty neighborhoods reduces violent criminal behavior by teens. Kling, Ludwig, and Katz (2005) exploited the same setting to deepen this relationship, noticing a gender difference and discussing several explanations for it. The offer to relocate to lower-poverty areas reduces arrests among female youth for violent and property crimes. The same offer produces different effects among boys: problem behaviors and property crime arrests increase while, at least in the short run, a reduction in arrests for violent crime is present. Long-run effects of MTO on youth outcomes after 10 to 15 years are reported to be like those reached after 4 to 7 years (Gennetian et al., 2012).

2.1.d Criminal inertia and other factors

Another root of criminal activity may be criminal hysteresis (i.e., inertia). As criminals pass through a learning-by-doing process, they can acquire significant criminal know-how, which in turn reduces the costs of committing criminal acts over time. From these considerations, the notion of criminal human capital emerged, viewing it as a substitute for legal human capital. Criminal human capital can indeed be accumulated and enhances an individual's opportunities in the crime market rather than in the legal labor market (Lochner, 2004; Mocan, Billups and Overland, 2005). Moreover, Buonanno and Leonida (2006) argued that people may tend to reduce their moral threshold after entering the crime industry, and past

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[4] Ludwig, Duncan, and Hirschfield (2001) provide a brief yet deep review of the literature on the topic.

[5] Looking at the variation in crime rates across cities, they observed an excess of criminality compared to what is predicted by measurable city characteristics, supporting the existence of neighborhood or peer effects.

incidences of crime in society may decrease the perceived probability of apprehension. Their results confirm that higher crime today is associated with higher crime tomorrow, as well as those from Fajnzylber, Lederman and Loayza (2002a, b) on violent crime.

Among other factors that can influence crime, Witte and Tauchen (1994) found that higher IQ and parochial school education are associated with lower criminal tendencies. Additionally, it seems that early childhood health and nutrition play a significant role in adolescent or adult criminal behavior (Olds et al., 1998; Olds, Sadler and Kitzman, 2007; Barr and Smith, 2023). Underage drinking (French and Maclean, 2006) and curfew ordinances (Kline, 2012) also negatively impact criminal behavior among youth.

2.1.e Education

The role of education on crime has been progressively examined too, suggesting that – if there is any effect – the social returns to education may even surpass the private returns individuals consider when making their educational choices, due to the significant negative externality of crime (Lochner and Moretti, 2004). In addition to the extensive literature on private returns to education,[6] studies have explored other social returns to education, such as whether the schooling of one worker raises the productivity and earnings of other workers around him (Heckman and Klenow, 1999; Acemoglu and Angrist, 2000; Moretti, 2004a, b) and examined the causal effect on mortality (Lleras-Muney, 2005). Given the potential wide-ranging social benefits of education, funding this field could help policymakers address several challenges at once, killing multiple birds with one stone.

Finally, it is worth mentioning that while education is frequently seen as a weapon to fight economic inequality, this viewpoint is not well supported by the research on the subject. Indeed, some recent studies challenge this

[6] See Deming (2022) and his references for a literature review.

perspective, suggesting that educational expansion policies may increase income inequality in the long run, while having no significant short-term effects (Makhlouf and Lalley, 2023). Previous analytical and empirical studies have also yielded varied results, offering a range of outcomes from theoretical models and mixed evidence.[7]

2.2 Relationship between education and crime[8]

2.2.a From theoretical models to empirical founding

Becker (1968) has pioneered the formalization of an economic approach to studying criminal behavior. In its model, as youths leave school, they face a trade-off between legal and illegal activities. Ehrlich (1973), Witte (1980), and Witte and Tauchen (1994) then examined engagement in crime as a decision related to the allocation of time. When individuals allocate (or are required to allocate) time to education (e.g., being at school), they are incapacitated from engaging in criminal activities outside of school.

With the introduction of dynamic aspects to better depict crime, several theoretical reasons through which education may negatively affect it have emerged in the literature over the years. Lochner and Moretti (2004) observed that, from an individual's perspective, as private returns to education increase through legitimate employment opportunities, the opportunity costs of engaging in illicit behavior also rise. Analogously, the potential time "lost" while being incarcerated becomes more costly due to those higher wages.

Moreover, education may also influence one's preferences regarding crime through external channels: Kling (1999) observed that among white-collar criminals, unlike blue-collar criminals, incarceration leads to substantial negative earnings effects. White-collar criminals earn 10-30% less after 5 to 8 years compared to those who were convicted at the same time but

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[7] For a solid literature review on the topic, see Makhlouf and Lalley (2023).

[8] Overall, Carr, Marie and Vujić (2023) provide a similar literature review on the relationship between education and crime.

were not incarcerated. General characteristics of the individual may also be affected by education, such as patience (Becker and Mulligan, 1997). This trait can lead more educated individuals to consider future punishments more carefully. In his summary of many studies from the psychological and neurological literature, Oreopoulos (2007) concluded that young people who drop out of school tend to be shortsighted and more concerned with the short-term costs of education (stress from tests, dull curricula, lost wages, etc.), instead of focusing on the long-term benefits of continuing their education. This line of literature also suggests that adolescents lack abstract reasoning skills and are more predisposed to risky behavior. Furthermore, Usher (1997) emphasizes that education perpetuates societal values, enculturates individuals to serve their communities and promotes virtues such as hard work and honesty.

However, as Lochner and Moretti (2004) pointed out, a negative correlation between education and crime may appear even without any causal effect of the former on the latter. Unobserved endogenous characteristics (such as local private return to crime outshining licit return or any other personal characteristics) may lead individuals to leave education behind and onset crime. Moreover, reverse causality may be an issue: being already successful in criminal activities might induce individuals to leave education behind as well.[9] Along the same line, the probability of detection and punishment or the sentence lengths handed out by judges may be reduced by education, as suggested by some evidence from Mustard (2001). Additionally, certain situations may result in a positive correlation between education and crime. For example, some white-collar crimes likely require higher levels of education[10]. Finally, the use of administrative data consisting of arrest and incarceration records may be problematic, since

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[9] Hjalmarsson (2008) provides some evidence that this is a valid concern, at least regarding crimes committed during the teenage years.

[10] Lochner's (2004) human capital model of crime implies that the relationship between education and white-collar crime should at least be less negative than that between "unskilled" crimes and may even be positive, as he empirically found (yet insignificant).

these measures are only observed for individuals who are caught. Following the line that more educated individuals are “better” criminals and thus less likely to be caught, convicted and/or incarcerated conditional on arrest, this can bias estimates of the relationship between education and administrative crime measures, as noted by Hjalmarsson, Holmlund and Lindquist (2015).

Despite not controlling for the potential endogeneity of schooling to crime, some studies have found an association between time spent in school and lower levels of criminal behavior (Farrington et al., 1986; Witte and Tauchen, 1994; Tauchen, Witte and Griesinger, 1994). Buonanno and Leonida (2009) also illustrate that education is negatively related to adult crime. On the other hand, the high school diploma alone does not appear to affect crime (Tauchen, Witte and Griesinger, 1994; Witte and Tauchen, 1994; Grogger, 1998).

More recently, numerous empirical studies have explored the causal nature of the relationship between education and crime leveraging various environmental settings. Initially, two main explanations emerged: the incapacitation effect in the short run (i.e., potential criminals are kept off the streets while they are in school) and the human capital effect[11] in the medium-long run, which involves increasing returns to legitimate work, as the labor market pays higher wages for higher productivity. These initial explanations led to the development of two distinct strands of crime economics research. Recently, the explanation for the latter effect diverged once again: the long-run effect of education on crime can arise either through the above-mentioned relative return to legitimate work (compared to the returns to criminal activities) or because short-run "direct" incapacitation exhibits state dependence that leads to dynamic incapacitation (Bell, Costa and Machin, 2022).

[11] This is not only “direct” human capital accumulation that leads to labor market return – which is a private return. It provides social returns too, affecting other individuals, as discussed in Section 2.1.e.

2.2.b Education and changes in the crime-age pattern

Since plenty of works utilize changes in compulsory school leaving (CSL) laws to investigate the causality between education and crime, it is worth examining the relationship between CSL laws reforms and crime-age pattern, which exhibits a secularly consistent empirical regularity. Indeed, research dating back to Quetelet (1831) has illustrated that criminal activity peaks when individuals reach their late teens.[12] Many scholars have since confirmed this pattern, observing the beginning of criminal activity in early adolescence and its high-water mark when most youth should still be enrolled in secondary school, and then declines sharply afterward (among others, Farrington et al., 1986; Levitt and Lochner, 2001; Sampson and Laub, 2003; Siennick and Osgood, 2008). In this context, Hirschi and Gottfredson (1983) conjectured the invariance of crime-age profiles over time and space. They stated that criminals may be identified by looking at their lack of self-control, a trait that develops before adolescence and persists throughout life. This would imply a flat crime-age profile, contradicting their conjecture. Instead, they later concluded that the opportunities for crime diminish over time as individuals' activities change with maturation (Gottfredson and Hirschi, 1990). In the same effort to explain the inverse U-shape of the crime-age profile, Sampson and Laub (2003, 2005) emphasized how events such as family, relationships, schooling and employment change as an individual ages. They characterize crime as a product of persistent individual differences and local life events. These life-cycle dynamics of crime generate that profile, with crime onset, specialization, and desistance occurring at distinct stages of life.[13]

Returning to changes in compulsory school leaving (CSL) laws, it is straightforward to see that once individuals reach the new, legally

[12] Lochner (2004) argues that white-collar crimes may peak way after teen years or even increase as individuals get older, in accordance with his human capital model. However, in the next paragraph, the crime-age profile considered is the general one.

[13] For a further dive into the topic, see Bell et al. (2022) and their references.

mandatory dropout age, the direct incapacitation effect will disappear – as they are free to now roam the streets. If education were solely responsible for this effect, then increasing the dropout age would impact the crime-age profile for teenagers younger than or equal to the new dropout age, with no significant effect on older criminals.[14] Conversely, the human capital effect would be supported if there was a considerable decrease in criminal activity among older individuals but a minimal effect on younger individuals below the new dropout age. The productivity-enhancing benefits of additional schooling – which lead to increased relative returns to legal activities – take time to manifest, making the impact more noticeable later in the crime-age profile rather than earlier.

It may be interesting to note that the concept of criminal capital is consistent with a medium-long effect of education on crime. Specifically, if compulsory school leaving (CSL) laws reforms extend schooling and thereby slow down the accumulation of criminal capital, this would affect the crime-age profile differently based on the underlying causal mechanism. If the only shift down in the crime-age profile was that in the early stages (consistently with the direct incapacitation effect), the erosion in criminal capital accumulated – relative to a counterfactual scenario without CSL laws reforms – would not influence the later stages of the crime-age profile, leading to a contradiction. Conversely, if the only reduction in the crime-age profile was in the late stage (consistently with the human capital effect), the reduction in criminal capital would have led to increased relative returns to legal activities as well. As mentioned in Section 2.2.a, in the early days of crime economics, engagement in crime was examined as a decision related to the allocation of time. This perspective is supported by the ideas

[14] Anderson (2014) argues that – despite not finding any evidence supporting this position – it is possible that criminal behavior may be merely pent up after an increase in the minimum dropout age while featuring an increase (rather than any effect at all) when individuals will later leave school.

of human and criminal capital, which follow the total time allocated to either education or crime.

Furthermore, Bell, Costa and Machin (2022) recently highlighted that in a dynamic framework, direct incapacitation may have an additional impact on future crime participation of an individual, regardless of any educational value associated with the incapacitation. As they illustrate with an example (p. 736): "Suppose being kept in school during the day prevents an individual from being on a street corner dealing drugs. This reduces arrests at the time but also potentially means that the individual leaves school without the criminal record they would otherwise have had. They now find it easier to pursue life as a law-abiding citizen." Lochner and Moretti (2004) anticipated this possibility of a dynamic incapacitation effect, suggesting that criminal behavior might exhibit strong state dependence, where the likelihood of committing a crime today is influenced by the amount of crime committed in the past.

2.2.c Incapacitation effect

Early studies of crime economics suggested the presence of an incapacitation effect, noting that working and attending school both significantly and nearly equally lower the likelihood of committing crimes. Indeed, the time spent on lawful activities – and not available for unlawful ones – may be responsible for the reduction in criminal behavior. (Farrington et al., 1986; Witte and Tauchen, 1994). Moreover, Kline (2012) found that imposing pinpoint youth curfews effectively lowers both violent and property crime: reducing the time spent – specifically, not spending certain hours – on street corners appears to be sufficient to decrease both violent and property crime. Anderson (2014) investigated the effect of different CSL laws across the US, noticing that raising the minimum dropout age significantly and negatively impacts property and violent crime arrest rates among 16- to 18-year-olds.[15] In a related context, Jacob and Lefgren (2003) initially looked at calendar year changes among US counties, followed by Luallen's (2006) work on the effect of teacher strikes. These two studies aimed to investigate whether changes in the requirement to be in school on a particular day have effects on crime. Both works show that property crimes committed by youngsters drop when school is in session, but violent juvenile crime rates increase on these same days. As the authors suggest, the incapacitation effect likely explains the property crime results, while evidence about violent crime findings may be due to a concentration effect – i.e. having many youngsters close to each other.

2.2.d Concentration effect, peers and school quality

As mentioned in Section 2.1.c, social circles play a crucial role in the criminal behavior of individuals. Consistently with the "general" neighborhood effect on crime, urban schools in high-poverty neighborhoods have high rates of violence and school dropout and struggle to retain effective teachers (among others, Lankford, Loeb and Wyckoff, 2002; Cook, Gottfredson and Na, 2010). On this line, Billings, Deming and Ross (2019) showed evidence for concentration effects in a setting of school resegregation in the US county of Charlotte-Mecklenburg, reporting also that these underprivileged youths are more likely to be arrested for committing crimes together – "partners in crime", quoting the authors. Billings, Deming and Rockoff (2014) leveraged the same setting to find that school resegregation led to an increase in racial inequality. Both whites and minorities score lower on high school exams when they are assigned to schools with more minority students, especially among high school students. Consistently with a step back on the education side, they found that rezoning led to a large and persistent increase in criminal activity among minority males.

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[15] Consistently with the studies mentioned right afore, Anderson (2014) himself argued that dropouts who seek employment may be less likely to commit crimes than those who leave school and have no interest in working. Notably, he also pointed out that other students may bear costs when an increase in the minimum dropout age forces more delinquents to remain in school.

Deming (2011), again from this setting, estimated the impact of attending a first-choice middle or high school on adult crime. He leveraged a setting where public school admission was determined by a lottery system. The years after enrollment in the preferred school are those which are responsible for the largest decline in crime. The impact is concentrated among high-risk youth, who commit about 50% less crime across several different outcome measures and scaling of crime by severity. Among middle school lottery winners, there was no overall reduction in crime, only a substitution toward property crime and away from violent crime. Changing peer groups seemed to be the most important mechanism for reductions in violent crime. On the other hand, there was no long-run reduction in criminal behavior. One potential explanation is that although changes in social settings can reduce concentration effects, persistent reductions in criminality are difficult to achieve without any human capital effect – nor dynamic incapacitation. Consistently with this observation, among high school lottery winners, there was an overall reduction in arrests and more evidence of increases in (nonpeer) school quality.[16] Exploiting the outcomes of that same lottery, Deming et al. (2014) found statistically significant increases in high school graduation, postsecondary attendance, and degree completion for students who win the lottery to attend their first choice. Gains on several measures of school quality strongly predict the impacts of choice. These results may be coherent with the negative effect of crime through the human capital mechanism.[17] Notably, a gender difference emerged: girls appeared to respond to attending a better school with higher grades and increases in college-preparatory course-taking, while boys did

[16] Huttunen et al. (2023) also found no evidence for peer effects during secondary education. As they worked in a Finnish setting, secondary education students ranged from 16 to 18 years old, virtually the same age as US high school students.

[17] Hanushek (2011) provides not only a literature review on the relationship between teacher quality and student achievement but also estimates the economic impact of improving teacher quality.

not. Therefore, as most criminals are male, the just-mentioned effect may not be present.

However, Arum and Beattie (1999) noticed earlier in the days that young men who enroll in secondary occupational coursework significantly reduce their likelihood of incarceration both overall and net of differences in the adult labor market – consistently with direct and dynamic incapacitation. High school student-teacher ratio and student composition also significantly affect an individual's risk of incarceration. Ladd and Sorensen (2017) found, among middle schools in North Carolina, large returns to teacher experience in terms of higher test scores and improvement in students' behavior, especially regarding absenteeism. As these results extend way beyond school years, teacher experience may affect criminal behavior not only because of direct incapacitation but also due to dynamic incapacitation or human capital effect. However, Rose, Schellenberg and Shem-Tov (2022) noted that, leveraging a North Carolina setting as well, the effects of teachers on criminal justice outcomes are "orthogonal" (as the authors put it) to their effects on academic achievement. Nevertheless, teachers who negatively affect suspensions and improve attendance reduce future arrests – in accordance with a direct incapacitation effect, as mentioned right above. Moreover, Jackson et al. (2020) found that Chicago high schools that fostered students' self-reported socioemotional development – by enhancing social well-being and encouraging hard work – were able to reduce school-based arrests and increase high school completion and college enrollment, even when accounting for their effect on test scores. Overall, these recent findings suggest that the development of non-cognitive skills might play a crucial role in the returns to education for crime – a topic that will be further explored in Section 2.3.b.

2.2.e Human capital and dynamic incapacitation effects

A – broader – strand of research focus asks the question of whether extra time spent in the education system induced by CSL laws changes has a longer-term effect on an individual's productivity. Emphasizing the underlying mechanism once more, additional education can brighten future labor market opportunities, therefore deterring those impacted individuals from pursuing a life of crime. Indeed, studies that can provide evidence of long-term benefits in crime reduction take advantage of schooling laws reforms that impacted those old enough to have completed their education. Among them, Lochner and Moretti (2004) found that in the US one extra year of schooling results in 0.10% and 0.37% reductions in the chance of incarceration for whites and blacks, respectively, and a reduction in both violent and property crime of 11–12%; Machin, Marie and Vujić (2011) found – in the UK – a significant, negative effect of education on property crimes and an insignificant effect on violent crimes.

Quite not of the same standpoint, Bell, Costa and Machin (2022) took advantage of a more general way of modeling the impact of school dropout age reforms on crime to show that they reshape crime-age profiles, rather than just shifting it down. While the negative effect is larger for those age groups who are directly constrained (i.e., incapacitated) by the reform, a significant negative effect is present even among those who are no longer directly constrained. As the authors report, this latter effect is interpreted as a dynamic incapacitation (rather than human capital accumulation) because further evidence indicates that these same reforms had at best quite mild effects on average educational attainment and wages, though they had somewhat more substantial effects on high school dropout rates.

In the footsteps of some previous works on the impact of Sweden's compulsory schooling reform of the 1950s on crime, Hjalmarsson, Holmlund and Lindquist (2015) utilized that setting to find that an additional year of schooling decreases the likelihood of a male conviction and prison sentence by 6.7% and 15.5%, respectively. However, as they point out, the magnitude is that large assuming that all the effect of the reform on crime is through increased schooling, yet it may be due to many other factors affected by that reform. Their results suggest that additional schooling has a causal impact on an individual's propensity to commit their first illegal act and on their propensity to commit offences which are severe enough to warrant imprisonment.

2.2.f Relative age within a grade

A strand of literature,[18] beginning with Angrist and Krueger (1991), has analyzed the effects of birth date on various outcomes. These studies consistently find that youths born just after the school-entry eligibility cut date (and thus relatively old for their grade) perform better academically than their younger classmates but are more likely to drop out before receiving their high school diplomas. Consequently, two potentially opposing human-capital mechanisms emerge in the delayed-entry "experiment," which are also respectively linked to two opposing crime-affecting mechanisms. On one hand, the better the academic performance, the higher the relative private returns to education compared to crime. On the other hand, the less time spent in school, the darker the chances of a criminal endeavor, as strongly stressed above. On this line, Cook and Kang (2016) and Landersø, Nielsen and Simonsen (2017) used variation in the age at school entry to determine whether crime aligns more closely with age or rather life-course factors. The former showed that a higher school starting age decreases juvenile delinquency but increases serious adult crime. However, their analysis is complicated by factors such as grade retention and compulsory school leaving age legislation, which creates a mechanical relationship between school starting age and the length of compulsory education[19] – which is proven to have a negative impact of crime, as previously discussed. On the other hand, Landersø, Nielsen and Simonsen (2017) leveraged exogenous variation in school starting age created by administrative rules in Denmark. They found that starting school

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[18] See Cook and Kang (2016) and their literature review.

[19] Landersø, Nielsen and Simonsen (2017) themselves addressed this issue concerning their colleagues' work in their paper. Indeed, within the same CSL laws environment, the negative effect on crime resulting from starting school later may be overshadowed by the positive effect on crime of spending fewer years in school.

at a later age reduces the likelihood of committing a crime during youth, but these effects begin to fade out in the twenties. This suggests that life-course variables, rather than age alone, can impact the crime-age profile. More evidence that age and criminal opportunity work together to influence the persistence of criminal conduct comes from the fact that a higher school starting age not only delays the initiation of criminal activity but also leads to a sustained decline in the number of crimes committed by males. These results appear to be consistent with the dynamic incapacitation effect. A higher school starting age for boys results in a large reduction in criminal charges up to the age of 19, with property crime being the main area of impact. Higher school starting ages for girls have the impact of delaying the initiation of criminal conduct; this is especially true for violent crimes. Additionally, they found that the crime reduction is not attributable to the relative age of classroom peers. Finally, Shepard (2024) used California criminal records to investigate whether being the youngest member of a cohort affected a person's likelihood of committing a crime. Although no persistent effect on the probability of arrest was found, the youngest student in a cohort seems to feature a higher risk of arrest for certain offenses at age 14 – the average transition age between middle and high school.

2.3 Non-compulsory education and crime

2.3.a Later education

While most of the crime economics literature focuses on compulsory education, Huttunen et al. (2023) studied the impact of secondary education on the criminal behavior of young men. Exploiting admission cut-offs in over-subscribed programs in Finland, they found results consistent with a dynamic incapacitation effect. Admission to any secondary school significantly reduced the propensity to commit crime, rather than merely delaying the onset of criminal activity, indicating that the direct incapacitation effect alone is insufficient to provide an explanation. Since the largest effects of admission on crime did not coincide with its effects on enrollment or labor market outcomes, the human capital effect is unlikely to be the primary factor. The absence of impact on crime from gaining access to a general track or to a selective secondary school rules out peer effects. The largest effect was observed in the years following the admission (when students were 16 to 18 years old). Furthermore, their finding suggests that access to secondary education primarily reduces minor crimes common among young individuals, rather than preventing the onset of criminal careers involving more serious crimes.

Furthermore, Nordin (2018) found that increasing the tertiary education eligibility rate decreases both property and violent crime substantially in Sweden, but these effects fade out when the marginal individual is less likely to be accepted into tertiary education. These results are surely consistent with an incapacitation effect, as the voluntary allocation of time to tertiary education – rather than to idleness – decreases an individual’s probability of committing a crime. However, young people with a high likelihood of deviant behavior often feature poor upper-secondary school grades and may therefore lack eligibility for tertiary education. Consequently, those who could benefit the most from escaping a criminogenic environment through tertiary education often lack the necessary eligibility to pursue it. Compared to other studies on the incapacitation effect, no concentration effects were reported on violent crime. Better yet, results suggest that tertiary attendance reduces violent crimes. The author suggests two possible explanations for this last evidence: it may be due either to the better peer group associated with tertiary education or to the spatial separation of potential victims and students, who congregate in different areas. Overall, combining this evidence with the earlier discussion presented in this work, the influence of peers appears to progress in a positive direction as students age.

2.3.b Early education

On the other hand of later education, early childhood education interventions (focused on quality and accessibility) may have a significant impact later in life (e.g., Chetty et al., 2011). Indeed, Cunha and Heckman (2008) showed that cognitive and (especially) non-cognitive skills at preschool ages are key determinants of later skill acquisition, behavior and adult outcomes. A variety of evidence suggests that noncognitive skills are much more closely related to impulsive and criminal behavior than cognitive skills and that each skill type can be influenced independently (Heckman, Stixrud, and Urzua, 2006; Heckman et al., 2010; Hill et al. 2011; Jackson, 2018). This lack of relationship between effects on cognitive measures and effects on adult criminal behavior in prior studies may not generalize to all cognitive measures, such as those used by Ladd, Muschkin, and Dodge (2014) to estimate North Carolina’s Smart Start program effects. However, Deming (2011) used the same test score to demonstrate the long-run effects of school choice on criminal behavior despite the gains in school quality generating no effect on test scores, suggesting that the crime effects are not operating through the cognitive channel in this context. Anders, Barr and Smith (2023) reported that, under very conservative assumptions, the estimated effects of Smart Start on criminal behavior are more than seven times greater than the effect implied by the third-grade test score estimates and the conditional correlation between test scores and crime. Furthermore, they stated that the effects of Head Start on educational attainment underpredict effects on criminal behavior by an even larger margin. Overall, these results suggest that early childhood education likely imparts important noncognitive skills that are not captured by test scores, and that cognitive measures alone may not be predictive of long-run effects on criminal behavior.

In the process of evaluating the social return on investments in early childhood education programs, positive externalities are crucial. Unlike labor market returns on improvements in human capital, these benefits largely accrue to individuals who are not directly affected by the programs and thus may not be considered in parents' participation decisions. According to Heckman et al. (2010), effects on crime account for 40–65 percent of the estimated benefits of such a program. Given the substantial burden that crime imposes, even modest reductions may justify significant subsidies for early childhood education.

Initially, three early childhood programs were examined: Perry Preschool, the Abecedarian Project and Head Start. Perry Preschool was a small-scale, high-intensity intervention. Although some evidence of a reduction in criminal behavior was found, researchers disagree regarding the presence of statistically significant effects (Heckman et al. 2010; Anderson, 2008). Heckman, Pinto and Savelyev (2013) decomposed the effects of Perry Preschool on eventual criminal behavior and showed that they operate entirely through a noncognitive channel (externalizing behavior). A randomized evaluation of a similar program, the Abecedarian Project, indicates either no overall effect of the program on crime (Campbell et al., 2002; Anderson, 2008) or noted robust effects only for participating girls (García, Heckman, and Ziff, 2019). Finally, Head Start, the largest early childhood education program in the US. It provides several wraparound services alongside education and serves poor families.[20] Deming (2009) found a generally positive impact, though smaller than that of Perry Preschool. Garces, Thomas and Currie (2002) previously noted positive effects, yet different between white and African American individuals.[21] Moreover, early childhood context and counterfactual care options have changed dramatically from the 1960s to the present day. Therefore, even

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[20] As Anders, Barr and Smith (2023) stated, US Office of Child Development 1968 reported that the median family income of children enrolled in Head Start was less than half that of all families in the United States. They also reference Gibbs, Ludwig, and Miller (2011) for a comprehensive review of the literature on the Head Start program.

[21] Anders, Barr and Smith (2023) listed a variety of issues present in those previous studies: small samples, self-reported crime data and sibling comparison approaches that may underestimate the effects of community-level program availability in the presence of spillovers.

being sure of the role of early childhood education in preventing criminal behavior in earlier periods, the same relationship may not exist in recent years.

To address these flaws and leveraging the less old Smart Start program, Anders, Barr and Smith (2023) used administrative crime data that has provided significant precision advantages over other existing work to find that improvements in early childhood education led to large (20 percent) reductions in subsequent criminal behavior. Such a steep decline outshines the reductions suggested by earlier studies' estimates of test score improvements. While the benefits generated account for a large portion of the costs of the education provided, their finding suggests that the effectiveness of additional early education funding may feature diminishing returns. Indeed, much smaller average effects were observed than those implied by several older studies of programs implemented in very high-poverty, underserved settings (such as Head Start), suggesting large potential gains from a more targeted approach.

3. Institutional setting and data

3.1 Overall considerations

The analysis will be conducted using data at the provincial level (NUTS-3) to ensure a comparable dataset for both education and crime while providing sufficient granularity. Indeed, Italian provinces have an average population of around 500,000 inhabitants and 80,000 students. However, it is important to note that provinces vary significantly in size. For instance, Lombardy alone accounts for one-sixth of the total population and has more than a million students.

Moreover, successive administrative reforms were conducted in Sardinia in recent years, causing both the number of provinces and their size to vary multiple times. For this reason, this region will not be included in the analysis. Therefore, the dataset is composed of 102 provinces.

3.2 Education

Data from the 2021/2022 school year can be used to provide a quick cross-level overview of Italy's compulsory education system.[22] Public schools dominate the educational environment, with private institutions only exceeding a tenth of the student population at the kindergarten level (Figures I and II). Class sizes remain consistent, averaging around twenty students per class. Gender balance is also maintained, as the female share just shallow of 50% reflects the overall demographics of youth in Italy. The proportion of foreign students is notable, especially among younger age groups (compatible with immigration trends and Italian citizenship regulations). Repetition rates are still significant at the high school level even if they declined after the COVID-19 epidemic (Figures III and IV).

Each spring, all Italian students in grades 2, 5, 8, 10 and 13 undergo standardized tests known as *Prove INVALSI*. These tests are designed by the INVALSI institute – *Istituto Nazionale per la Valutazione del Sistema Educativo di Istruzione e di Formazione* – with the primary aim of measuring the quality of education and student learning outcomes nationwide. They provide schools and policymakers with data for self-evaluations and overall decision-making, respectively. INVALSI also participates in international assessments, such as PISA and TIMSS, in collaboration with organizations like the OECD and IEA, enabling Italy to benchmark its educational performance with that of other countries.[23]

The tests currently cover three subjects – Italian, mathematics and English – and are composed of multiple-choice questions. They are administered at different grade levels: second and fifth grades (primary school), eighth grade (middle school), tenth and thirteenth grades (second and final years of high school)[24]. The format of the tests varies by grade level. For students

[22] Data source: http://dati.istat.it//Index.aspx?QueryId=65615#

[23] More information may be found at https://www.invalsi.it/invalsi/index.php.

[24] In Italy, high school spans 5 years, from grade 9 to 13, whereas a bachelor's degree lasts only 3 years.

in primary school, the tests are paper-based, while for older students in middle and high school, they are computer-based testing (CBT). The questions are designed to assess not only factual knowledge but also critical thinking and problem-solving skills.

For each province, the dataset contains the grade level, subject, average score, standard deviation and the proportion of students who attended the test compared to the expected attendance.[25] Since teachers may have helped students cheat, average scores are constantly corrected for cheating. Given that several years' worth of observations is available, the data are pooled into a panel. However, some issues arise due to gaps in the panel, as the legislation governing the tests has changed multiple times. Scores for Italian and mathematics have been available since the school year 2012/2013 for grades 2, 5, 8 and 10. English reading and listening abilities have been tested since 2017/2018 in grades 5 and 8. Tests in grade 13 appeared in 2018/2019. Moreover, due to the COVID-19 pandemic, tests were not conducted during the school year 2019/2020; in the following year (2020/2021) assessments were not conducted for grade 10. Additional gaps in the dataset are due to the format of tests. CBT was introduced in 2017/2018. Initially implemented for grades 8 and 10, it was later extended to grade 13 when tests for this grade were introduced the following school year. The last year of observation in the dataset is 2022/2023. For CBT (but also for paper-based tests) average score and standard deviation are only available as the WLE average ability parameter of the Rasch model, whereas for paper-based tests they are expressed also as percentage points of correct answers.

The Rasch model – which is a part of the Item Response Theory (IRT) family – is used to measure latent traits, such as abilities or attitudes, based on responses to test items or survey questions. In this model, the probability

[25] Data source: https://serviziostatistico.invalsi.it/invalsi_ss_data/dati-provinciali-di-popolazione/

of answering a question correctly reflects the differences between two parameters measured on the same scale: the ability parameter of the respondent and the difficulty parameter of the question. It is important to note that the relationship between these two parameters and the probability of answering correctly is not linear. Indeed, it follows a logit scale. As a result, the most significant increase in the probability of answering correctly occurs when this probability is around 50%. In contrast, even substantial relative gains in ability will have minimal impact on the probability of answering correctly when it is already close to 0 or 1.

The mathematics test results for each province in the school year 2022/2023 are shown in Figures V and VI, expressed as WLE average ability parameter serving. In primary school, no territorial trends appear, while a gap between Northern and Southern Italy widens through middle and high school. This pattern is consistent in INVALSI data across years and subjects, aligning with the broader economic landscape in Italy.

Additionally, the share of students across five different skill levels has been available since the 2017/2018 school year. These levels were provided for grades 8 and 10 in Italian and mathematics and for grades 5 and 8 in English. Starting the following year, this statistic has also been computed for grade 13.

3.2 Crime

Data from 2017 convictions can be used to provide a quick overview of crime in Italy.[26] The top ten most common felonies combine for more than half of the total. Among them, major property crimes combined (theft, fencing, home burglary and snatching, fraud) represent the greatest concern, although the single most common felony is drug-related crime (including both production and trafficking). Resisting a public officer and culpable injuries fall just a bit behind. Finally, there are violations of family

[26] Data source: http://dati.istat.it/Index.aspx?QueryId=25232

support obligations and evasion from custody. Regarding more serious crimes, manslaughter rates have been on a downward trend for decades, especially for men.[27] Despite Italy's reputation, mafia-related crimes represent all but a relevant share. On the other hand, threats and extortion are less rare. Overall, female crime in Italy is virtually negligible when compared to the amount of male crime. The felony in which the gap is closer – females commit half of the males – is insult, which is a minor crime, at least concerning numbers.

The shape of the crime-age pattern is consistent with its secular regularity. Criminal behavior peaks between the ages of 18 and 24, especially among the most common felonies (figure VII). Crime in the early teens (16-17 range) is not that spread.[28] On the other hand, the overall profile does not feature a sharp decline until the late 30's (figure VIII). This is due to the presence of certain common violations that do not pertain to youths, such as those related to family support obligations or social security. Fencing, fraud and evasion from custody do not feature a steep decrease, too.

Moving to the dataset, administrative data from the Department of Public Safety of the Italian Ministry of Interior will be used, consisting of the number of crimes reported to the Judicial Authority by the Police Force.[29] For each province, either the numbers of crimes or criminals are available for a broad range of felonies – 77 categories in total – in the period 2016-2023. These numbers are then linked with population data to compute the

[27] Data source: https://www.istat.it/statistiche-per-temi/focus/violenza-sulle-donne/il-fenomeno/omicidi-di-donne/

[28] This fact is coherent with a substantial incapacitation effect, as high school lasts 5 years. In fact, despite a non-negligible share of dropouts (see for example http://dati.istat.it/Index.aspx?DataSetCode=DCCV_ESL_UNT2020), potential offenders may remain in school due to sunk cost fallacy. Indeed, high schools are selected by students during grade 8 and will be the same until grade 13, though only grades 9 and 10 are compulsory and students may always move between high schools. Moreover, Huttunen et al. (2023) found similar evidence in Finland.

[29] Data source: https://ucs.interno.gov.it/ucs/contenuti/Numero_dei_delitti_denunciati_all_autorit_agrave_giudiziaria_dalle_forze_di_polizia_int_00062-7730889.htm

number of crimes (or criminals) per 100,000 inhabitants.[30] Overall, no significant geographical trends appear to be present in either the number of crimes or the number of criminals. As certain provinces feature greater numbers for a given violation, a visual representation of the phenomenon results in a "patchy" map (figure IX).

4. Empirical Strategy

WLE average ability parameter will serve as the main instrumental variable for education, while the main instrument for crime will be the total number of criminals per 100,000 inhabitants. Several other variables will also be assessed.

Since there is no evident "natural experiment" or clear natural threshold when plotting education against crime (figure X), a regression discontinuity design cannot be meaningfully conducted. The first relationship explored in the regression analysis is the most basic:

$$\mathrm{Crime}_{\mathbf{it}} = \boldsymbol{\alpha} + \boldsymbol{\beta}\,\mathrm{Edu}_{\mathbf{it}} + \boldsymbol{\epsilon}_{\mathbf{it}} \tag{1}$$

Crime_{it} is the outcome variable (number of crime or criminals per capita) for province i in year t. Edu_{it} is the independent variable for education (e.g., WLE average ability score or skill level).[31] Finally, ϵ_{it} is the error term. As discussed in Section 2.2, a negative value of β is expected. However, R^2 is expected to be relatively close to 0, as many factors other than education do affect crime – as mentioned in Section 2.1. Indeed, the value of α may be not only significantly positive but also substantial in magnitude. Hausman test is conducted to check for the presence of a fixed effect across provinces. Following this, fixed effects for time are added to the relationship to control for temporal variations in the relationship:

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[30] Data source: https://esploradati.istat.it/databrowser/#/it/dw/categories/IT1,POP,1.0/POP_INTCENSPOP/DCIS_RICPOPRES2011/IT1,164_164_DF_DCIS_RICPOPRES2011_1,1.0

[31] In this framework, school years are labeled as t-1/t (i.e. the 2022/2023 school year is represented as t=2023). Consequently, crime data for a given year is associated with the tests conducted in the spring of that same year.

$$\mathrm{Crime}_{\mathbf{it}} = \boldsymbol{\alpha} + \boldsymbol{\beta}\,\mathrm{Edu}_{\mathbf{it}} + \boldsymbol{\eta}_{\mathbf{t}} + \boldsymbol{\epsilon}_{\mathbf{it}} \qquad (2)$$

A general crime reduction through time is expected to be captured. Time fixed effect is estimated via Least Squares Dummy Variables. Finally, to further investigate the significance and magnitude of the effect of education on crime, it is necessary to account for additional factors using control variables, in accordance with Section 2.1:

$$\mathrm{Crime}_{\mathbf{i}} = \boldsymbol{\alpha} + \boldsymbol{\beta}\,\mathrm{Edu}_{\mathbf{i,n}} + \underline{\boldsymbol{\delta}}\,\mathbf{X}_{it} + \boldsymbol{\eta}_{\mathbf{t}} + \boldsymbol{\epsilon}_{\mathbf{it}} \qquad (3)$$

The matrix X_{it} includes, among others, the share of people in poverty, occupation and inactivity rates.[32]

The analysis will be conducted for each grade and subject using panel data. Equations above will be estimated using the population of each province (in the given year) as weights for WLS regression. Pearson test is conducted to check for the presence of endogeneity and multicollinearity. To further keep the latter issue under control, its severity is measured by computing the variance inflation factor (VIF).

5. Results

Estimates of (1) and (2) are presented in figures XI and XII, using the total number of criminals and the WLE average ability score as instruments for crime and education, respectively. The number of observations in the panel varies depending on data availability (see Section 3). The maximum number of observations is n=714 – as 102 provinces are observed from 2016 to 2023, excluding 2020. Due to the gap caused by the COVID-19 pandemic, the panel is consistently unbalanced. While the R^2 values peak at around 10%, test scores from as early as grade 5 seem to influence crime rates, with the most significant effects observed in Italian test scores from grades

[32] Poverty rate and NEET data are available only at the regional level. Occupation and inactivity rates data from ISTAT are also available, yet with gaps. Data source: https://esploradati.istat.it/databrowser/#/it/dw/categories/IT1,Z0500LAB,1.0/LAB_OFFER. Among other factors that affect crime – see Section 2.1 – and thus eligible as control variables, abortion-related data are available only at the regional level, too.

8 and 10. Specifically, a point gained in the WLE average ability score is associated with a reduction of more than 15 criminals. The results of the Hausman test support the inclusion of fixed effects for provinces, which aligns with the "patchy" nature of crime distribution across Italy (figure IX).

On one hand, the results remain consistent when other instruments for education are used, such as skill levels (see Section 3). A larger proportion of students in a relatively high skill level is negatively associated with the number of criminals, whereas for a larger proportion in the bottom skill level, the association is positive. On the other hand, the number of crimes – rather than the number of criminals – does not appear to be negatively linked with education. This evidence may be because a large proportion of crimes are committed by a small subset of habitual offenders – as mentioned in the Introduction – whose choices are not linked with educational outcomes.

When controlling for the fixed effect of time, the relationship between education and crime appears even in earlier grades. Additionally, its magnitude is greater. Furthermore, the fixed effects for time – estimated using Least Squares Dummy Variables – appear to be negatively significant ($p<0.01$ and often $p<10^{-3}$) for post-pandemic years. The intercept for these years is roughly 95% of the intercept of the base year, which varies according to the grade and subject of the test. Endogeneity is always absent, as the Pearson test confirms no significant – weighted – correlation between residuals and independent variables (the null hypothesis holds). Regarding multicollinearity, although the null of the Pearson test is often rejected, the VIF for education never features alarming values, frequently being close to 1.

In the estimation of (3) the more control variables that are included, the more serious the problem of multicollinearity gets. These controls exhibit not only a correlation with each other but also a stronger one with education. This problem is highlighted by examining weighted correlations; for instance, the correlation between the WLE average ability score and the proportion of individuals in poverty (using 2022 data) remains below -0.80 for most of the older grades.

Willing to control for the role of criminal inertia, multicollinearity arises once again. The weighted correlation between criminal data from the previous year and current education is significantly negative across all grades and subjects (the null hypothesis of the Pearson test is consistently rejected). Nevertheless, the positive relationship between current and past year crime (weighted correlation for the number of criminals always above 0.40 and often above 0.80, always above 0.98 for the number of crimes) may be a clue of criminal inertia – which was also found by Buonanno and Leonida (2006) in a similar setting. Further econometric analysis is needed to address its presence with more recent data as, among others, it may explain the persistence of differences between provinces.

A major limitation of this analysis is that crime data covers the entire population rather than being specific to youths – like in Nordin (2018) – as the modal range of criminality (18-24) does not incorporate a major proportion of criminals. Indeed, not only crime is far from absent in other age groups but also youth represent only a small fraction of the overall population in Italy. Therefore, other exogenous and unknown factors could potentially skew the results. Overall, even though a relationship between education and crime is evident in this context, further investigation is required to determine its magnitude and causal nature.

6. Conclusion

After a literature review on the negative relationship between education and crime, this analysis of the Italian context yields consistent findings. Specifically, higher educational test scores are linked to lower numbers of criminals in Italian provinces. Provincial fixed effects appear to be in place, alongside a general falling trend regarding the number of criminals. The negative association persists even when using alternative measures of educational attainment, such as skill levels among test takers. These results align with those from Buonanno and Leonida (2006, 2009) in a similar setting. However, the relationship between education and the total number of crimes is less clear, possibly because a small subset of less sensitive-to-education criminals is responsible for a sizable portion of offenses. Regarding endogeneity and multicollinearity, the former never appears to be in place, while the latter arises significantly when more controls (in accordance with existing literature) are included in the model.

A major limitation of this work is that the crime data on which the analysis relied cover the entire population, rather than youth crime, with the possibility that exogenous factors related to older criminals could skew the results. Moreover, despite the identification of statistical relationships, further investigation is needed to explore the causal mechanisms and to evaluate the impact of other variables, such as socioeconomic ones, on the observed patterns. For instance, policymakers may affect both educational outcomes and crime rates through welfare interventions. Furthermore, the role of criminal inertia is also to be deepened. Lastly, it would be essential to translate the results into economic terms rather than purely statistical ones, in line with recent developments in the literature, to give policymakers more useful insights.

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8. Figures

		Schools (with %)				
	Grade	Public		Private		Total
Kindergarten		15,164	68.1	7,119	31.9	22,283
Primary school	1-5	15,273	91.7	1,380	8.3	16,653
Middle school	6-8	7,411	92.1	639	7.9	8,050
High school	9-13	5,277	76.7	1,602	23.3	6,879
Total	K-13	43,125	80.1	10,740	19.9	53,865

Figure I illustrates the number of schooling institutions in Italy across levels (school year 2021/2022). ISTAT data.

	Grade	Students (with %)				
		Public		Private		Total
Kindergarten		958,483	72.6	360,973	27.4	1,319,456
Primary school	1-5	2,365,146	93.6	161,586	6.4	2,526,732
Middle school	6-8	1,618,507	95.9	68,679	4.1	1,687,186
High school	9-13	2,605,146	95.5	122,491	4.5	2,727,637
Total	K-13	7,547,282	91.4	713,729	8.6	8,261,011

Figure II illustrates the number of students in Italy across levels (school year 2021/2022). ISTAT data.

	Grade	Students per class			Repeating students (%)		
		Public	Private	Total	Public	Private	Total
Kindergarten		20.1	19.5	19.9	..	..	..
Primary school	1-5	18.0	18.2	18.0	0.003	0.003	0.003
Middle school	6-8	20.2	20.8	20.2	0.019	0.005	0.018
High school	9-13	20.4	14.2	20.0	0.066	0.068	0.066
Total	K-13	19.6	18.4	19.4	0.028	0.013	0.027

Figure III illustrates the number of students per class and the share of repeating students in Italy across levels (school year 2021/2022). ISTAT data.

	Grade	Foreign students (%)			Female students (%)		
		Public	Private	Total	Public	Private	Total
Kindergarten		13.3	7.5	11.7	48.0	48.1	48.1
Primary school	1-5	12.9	4.5	12.4	48.4	48.9	48.5
Middle school	6-8	11.5	3.7	11.2	48.3	47.5	48.3
High school	9-13	8.1	4.2	8.0	48.9	43.2	48.7
Total	K-13	11.0	5.9	10.6	48.5	47.4	48.5

Figure IV illustrates the share of foreign and female students in Italy across levels (school year 2021/2022). ISTAT data.

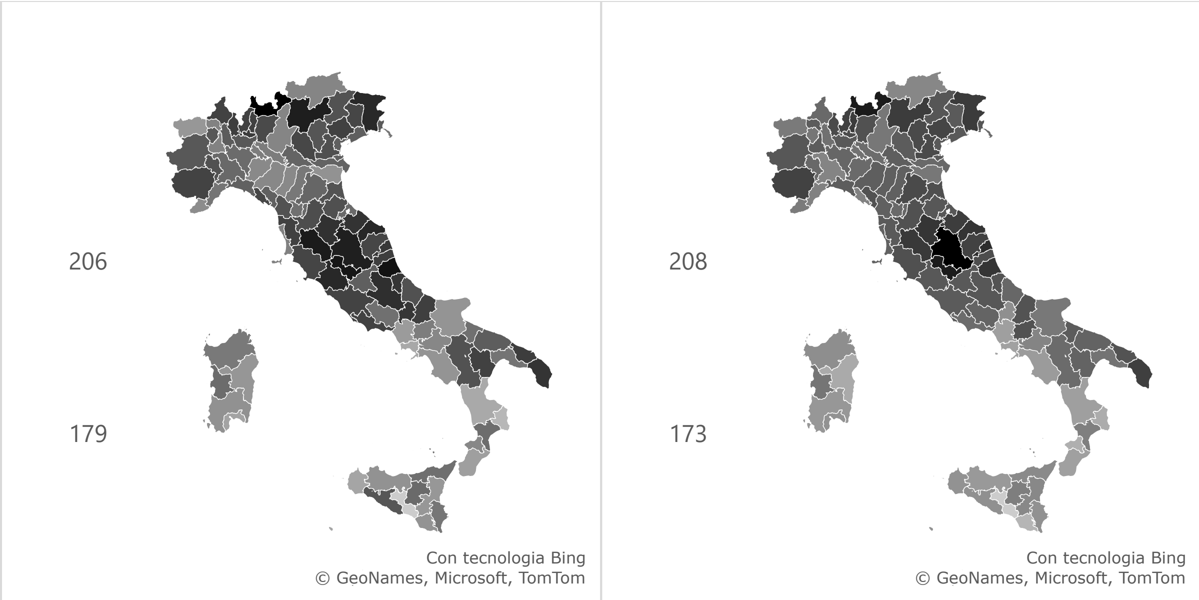

Figure V illustrates the WLE average ability score in mathematics for grade 2 and grade 5 students in school year 2022/2023. INVALSI data.

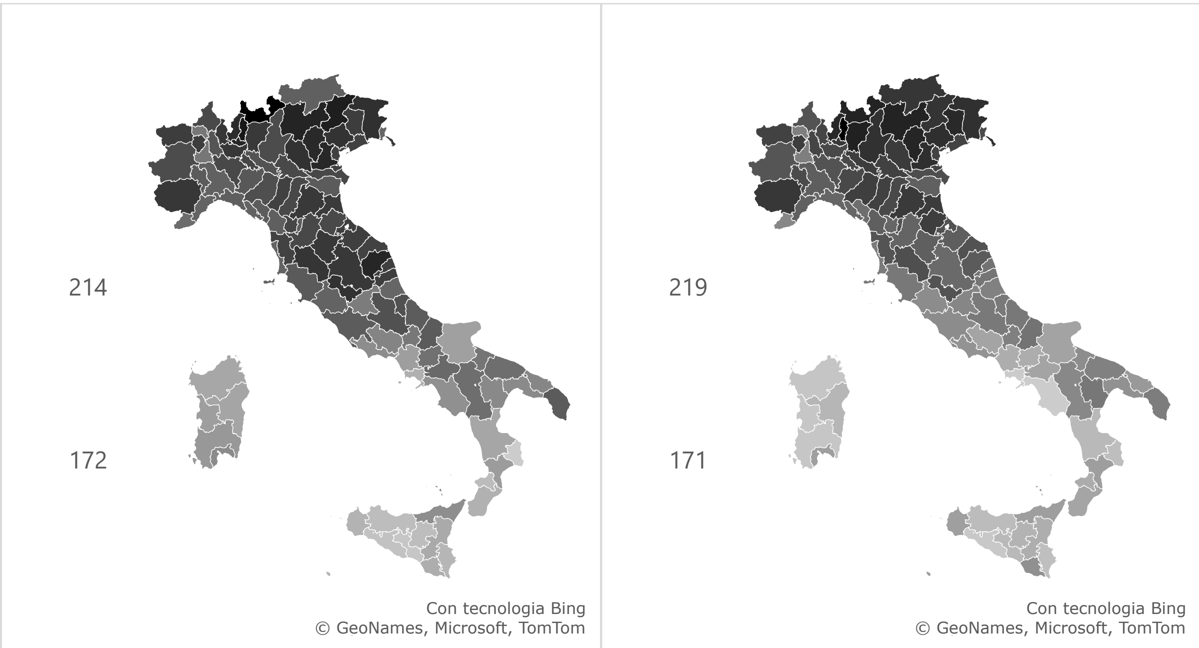

Figure VI illustrates the WLE average ability score in mathematics for grade 8 and grade 13 students in school year 2022/2023. INVALSI data.

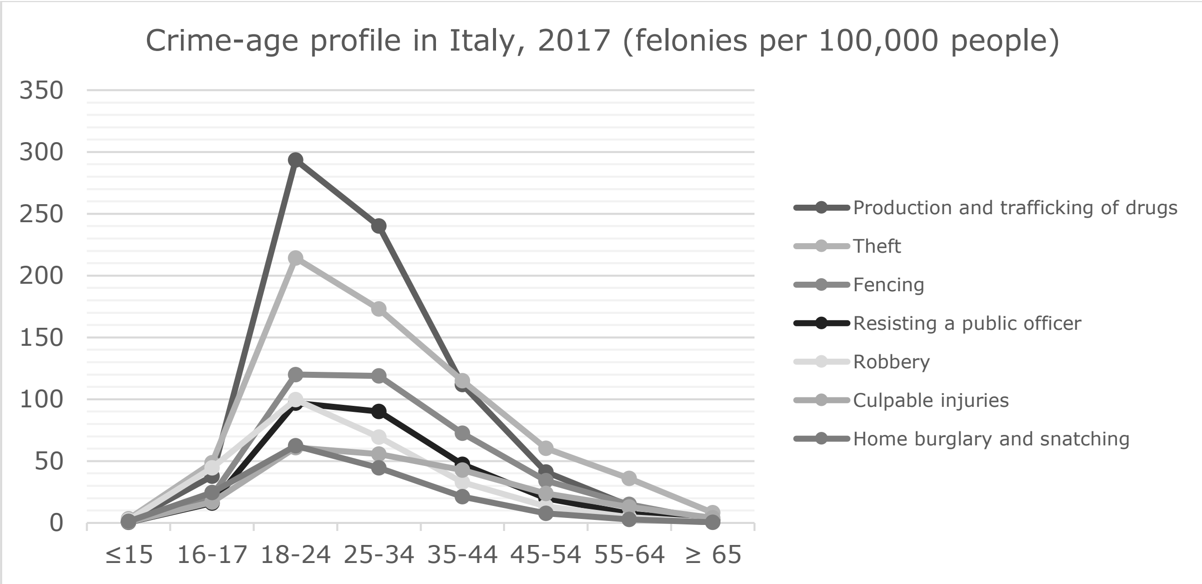

Figure VII represents the male crime-age profile in Italy in 2017 for most common felonies. Female crime is negligible in size compared to male crime. Instrument for felonies is irrevocable conviction. Elaboration from ISTAT data.

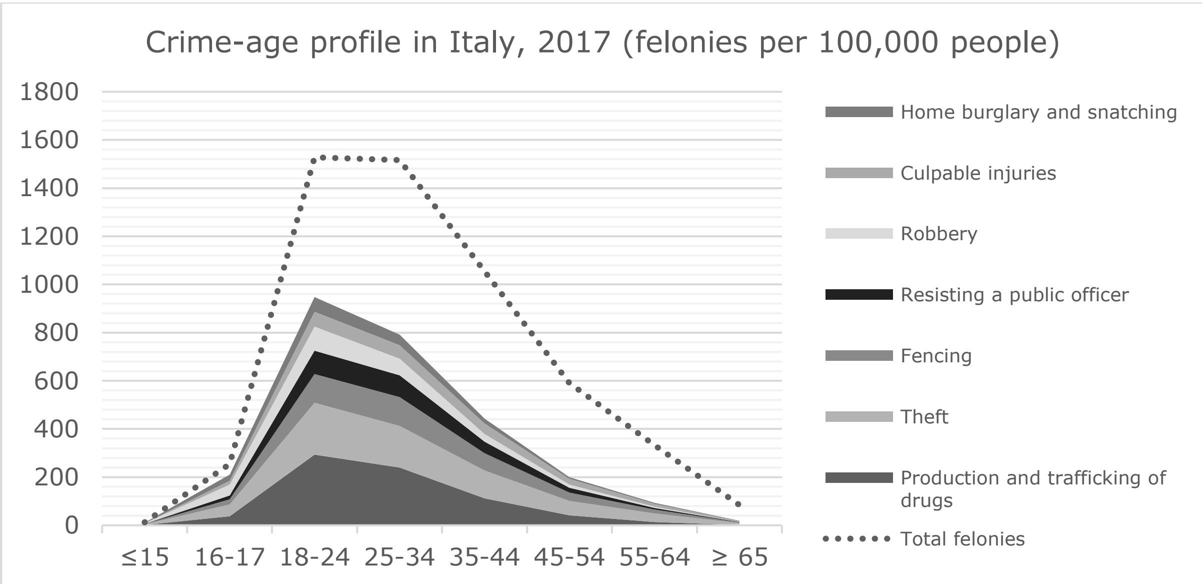

Figure VIII represents the male crime-age profile in Italy in 2017. Most common felonies add up to roughly half of total. Female crime is negligible in size compared to male crime. Instrument for felonies is irrevocable conviction. Elaboration from ISTAT data.

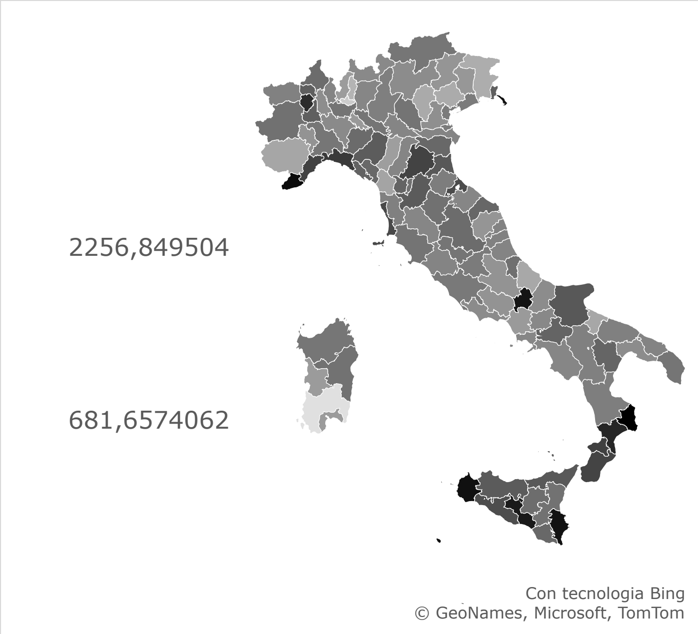

Figure IX illustrates the total number of criminals per 100,000 inhabitants in 2023. Italian Minister of Interior, Department of Public Safety data.

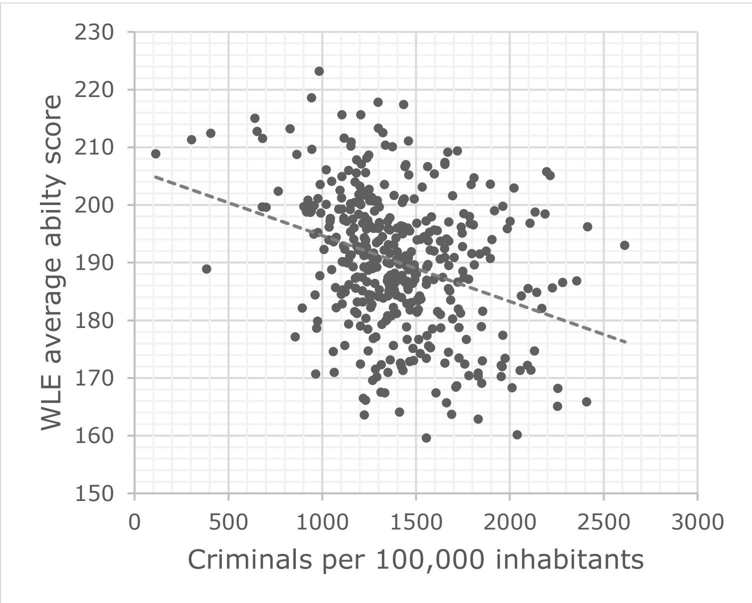

Figure X shows WLE average ability score in Italian for grade 13 students (school years 2018/2019, 2020/2021, 2021/2022, 2022/2023) scattered against number of criminals per capita (years 2019, 2021, 2022, 2023). Outliers (±2 SD in crime) are removed. The pattern is consistent across years and test subject. INVALSI and Italian Minister of Interior, Department of Public Safety data.

	Intercept		Education		F		R^2	n	H (df=1)	
Italian										
Grade 2	2561	(491)	-5.674	(2.408)	5.553		0.008	714	16.483	***
Grade 5	3793	(482)	-11.868	(2.394)	24.575	***	0.033	714	35.407	***
Grade 8	5320	(448)	-19.864	(2.273)	76.353	***	0.097	714	19.596	***
Grade 10	4467	(374)	-15.510	(1.896)	69.945	***	0.099	612	36.354	***
Grade 13	2897	(371)	-8.007	(1.948)	16.895	***	0.040	408	19.342	***
Mathematics										
Grade 2	2131	(487)	-3.564	(2.390)	2.224		0.003	714	10.551	**
Grade 5	2670	(485)	-6.275	(2.404)	6.811	**	0.010	714	34.642	***
Grade 8	3886	(314)	-12.627	(1.595)	62.652	***	0.081	714	28.918	***
Grade 10	3932	(303)	-12.737	(1.527)	69.604	***	0.102	612	24.757	***
Grade 13	2868	(359)	-7.651	(1.835)	17.381	**	0.041	408	16.890	***
English (L)										
Grade 5	3021	(587)	-8.078	(2.877)	7.884	**	0.019	408	19.506	***
Grade 8	3314	(302)	-9.463	(1.478)	40.991	***	0.075	510	0.000	
Grade 13	2930	(322)	-7.634	(1.575)	23.505	***	0.055	408	6.454	*
English (R)										
Grade 5	2841	(635)	-7.308	(3.161)	5.347	*	0.013	408	29.680	***
Grade 8	4188	(396)	-13.639	(1.927)	50.104	***	0.090	510	0.025	
Grade 13	3632	(450)	-11.211	(2.232)	25.220	***	0.059	408	0.092	

Figure XI illustrates estimated effects of education on crime. Instrument for education is WLE average ability score in each subject and grade (data from INVALSI). Instrument for crime is the number of criminals per 100,000 inhabitants (data from Italian Minister of Interior, Department of Public Safety). Last column shows the statistics for the Hausman test to signal the presence of province fixed effect. Despite low values of R^2, effect is always significant since grade 5.

	Intercept		Education		F		VIF	n	Controls
Italian									
Grade 2	5051	(750)	-17.679	(3.691)	4.350	***	2.412	714	22, 23
Grade 5	6050	(611)	-22.868	(3.041)	9.201	***	1.688	714	23
Grade 8	5710	(467)	-21.820	(2.391)	13.057	***	1.119	714	
Grade 10	5136	(391)	-18.677	(1.977)	16.064	***	1.060	612	22, 23
Grade 13	3607	(433)	-10.978	(2.154)	6.870	***	1.244	408	21, 22, 23
Mathematics									
Grade 2	4955	(758)	-17.176	(3.728)	4.103	***	2.503	714	22, 23
Grade 5	6329	(722)	-23.830	(3.532)	7.609	***	2.279	714	21, 22, 23
Grade 8	4233	(322)	-14.232	(1.646)	11.828	***	1.084	714	23
Grade 10	4192	(309)	-13.878	(1.559)	14.385	***	1.127	612	22, 23
Grade 13	3316	(400)	-9.354	(1.951)	6.118	***	1.142	408	21, 23
English (L)									
Grade 5	3316	(635)	-9.193	(3.063)	2.614	*	1.132	408	
Grade 8	3322	(310)	-9.510	(1.544)	8.178	***	1.083	510	
Grade 13	2933	(326)	-7.551	(1.606)	5.900	***	1.033	408	
English (R)									
Grade 5	3941	(799)	-12.195	(3.851)	2.869	*	1.496	408	21, 23
Grade 8	4226	(410)	-13.881	(2.021)	10.036	***	1.092	510	
Grade 13	3777	(467)	-11.628	(2.287)	6.836	***	1.048	408	

Figure XII illustrates estimated effects of education on crime controlled for time effect. Instrument for education is WLE average ability score in each subject and grade (data from

INVALSI). Instrument for crime is the number of criminals per 100,000 inhabitants (data from Italian Minister of Interior, Department of Public Safety). Last column indicates which years of observation features a significant reduction (in the intercept) compared to the first year of observation. Effect of education on crime is always significant.